

Hot Spot

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Time Limit: 10 s Memory Limit: 128 MB
Submission: 35 AC: 11 Score: 99.47

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Codes

AI Code Tips

Description

Hot Spot is a single player game played on a 4 by 4 game board. The purpose of the game is to move a red robot from its current location on the board to the top left corner. The game board may also contain green and blue robots. Each square of the game board can be occupied by no more than one robot at any time.

In one step, you can choose a robot, and move in one of the following two ways:

1. If two robots are adjacent horizontally or vertically (but not diagonally), one of them may jump over the other to the immediately adjacent square, provided that square is empty. For example, if robot a is immediately to the left of robot b, robot a may jump to the square immediately to the right of robot b.
2. If three robots are adjacent in a line (again not diagonally), one of them may jump over the other two, provided the destination square is empty. For example, if robot b is immediately to the right of robot a and robot c is immediately to the right of robot b, robot a may jump to the square immediately to the right of robot c.

Notice that you can not move a robot to a adjacent empty cell. The only way to move a robot is applying one of the rules mentioned above.

Every jump only changes the positions of the existing robots; robots are never removed from or added to the game board.

And there is another strange rule that a blue robot is never allowed to be adjacent horizontally or vertically to another blue robot or to the red robot.

Given the initial configuration of the game board, determine the minimum number of jumps required to move the red robot into the top left corner.

Input

The input specifies the initial position of the board using four lines, each containing four characters. Each character may be either R, indicating the red robot, B, indicating a blue robot, G, indicating a green robot, or a period (.), indicating an empty square. The squares of the map will not exceeds 100.

Output

Output a single line containing a single integer, the minimum number of jumps required for the red robot to reach the top left square of the game board.

Samples

input	Copy
<pre>.GR.</pre>	
output	Copy
<pre>1</pre>	

input	Copy
<pre>.B.. .GR. G... .G.G</pre>	
output	Copy
<pre>8</pre>	

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